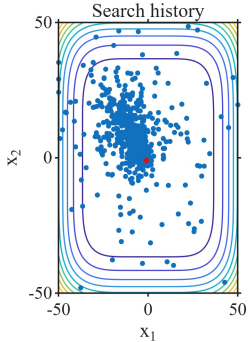
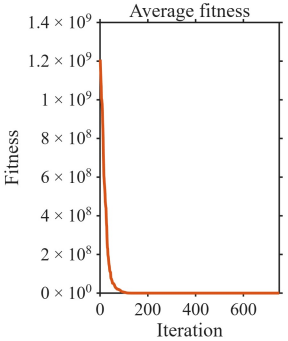
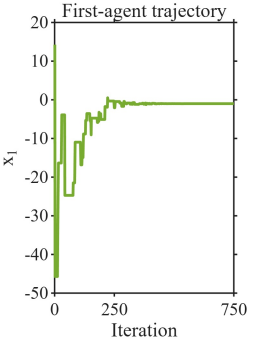
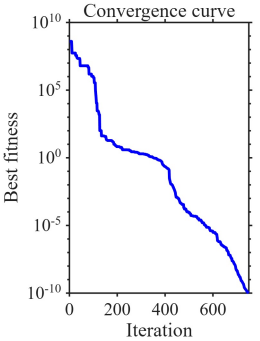

F12
Fitness
140
120
100
80
60
40
20
10
0
-10
x2
-10
0
10
x1


Search history
x2
50
0
-50
-50
0
50
x1


Average fitness
Fitness
1.4 × 10^9
1.2 × 10^9
1 × 10^9
8 × 10^8
6 × 10^8
4 × 10^8
2 × 10^8
0 × 10^0
0
200
400
600
Iteration


First-agent trajectory
x1
20
10
0
-10
-20
-30
-40
-50
0
250
750
Iteration


Convergence curve
Best fitness
10^10
10^5
10^0
10^-5
10^-10
0
200
400
600
Iteration